

AI-Powered Campus Food Intelligence Agentic Workflow

Complete End-to-End ServiceNow Implementation Guide

ServiceNow | AI Agents | Agentic Workflows | Now Assist | Generative AI

1. Project Overview

This project demonstrates how ServiceNow AI capabilities can be used to move from a simple AI assistant toward an agentic enterprise workflow. The workflow understands a natural-language request, identifies the required information, delegates the task to a Food Management Intelligence Agent, uses relevant analytics tools, combines the results, and produces evidence-based operational insights.

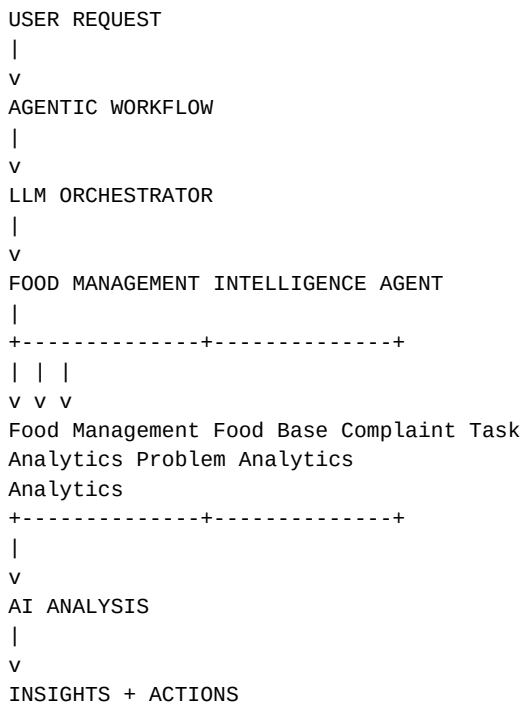
Core idea: User Request → Agentic Workflow → AI Agent → Relevant Tools → AI Analysis → Insights → Recommendations

2. Problem Statement

Campus food management information can be distributed across complaints, recurring food problems, complaint tasks, priorities, mess halls, meal types, student impact, and resolution status. Manually checking several sources to understand the current situation can be time-consuming.

The goal is to provide an AI-powered workflow that can understand a management request and coordinate the required analysis automatically, while keeping recommendations grounded in available ServiceNow data.

3. Solution Architecture



4. Prerequisites

- ServiceNow instance with the required AI Agent / Agentic Workflow capabilities enabled.
- Access to the relevant AI configuration experience and required roles.

- Campus food management tables and realistic test records.
- Analytics/data retrieval capabilities that can be exposed as AI tools.
- Now Assist / Generative AI capability where required by the selected ServiceNow feature.

5. Prepare the Application Data

Before creating the AI layer, make sure the application contains meaningful data.

Food Complaint

- Complaint number
- Use case
- Mess hall
- Meal type
- Priority
- Student count
- Description
- Created date

Food Base Problem

- Problem number
- Problem category
- Use case
- Mess hall
- Frequency
- Related complaints

Complaint Task

- Task number
- Complaint reference
- Current priority
- Assignment group
- Resolution status
- AI summary
- Priority reason
- Recommended actions
- Student count

6. Create the AI Tools

Tool 1: Get Food Management Analytics

Purpose: Provides overall food management statistics and operational information.

Recommended output: total complaints, complaints by use case, mess hall, meal type, priority distribution, student impact, and relevant trends.

Tool description: Retrieves overall campus food management analytics including complaint counts, priorities, mess halls, meal types, use cases, student impact, and relevant operational trends.

Tool 2: Food Base Problem Analytics

Purpose: Identifies recurring or categorized food problems.

Recommended output: base problem categories, frequency, affected mess halls, related use cases, student impact, and recurring patterns.

Tool description: Analyzes food base problem records to identify recurring food-related issues, problem categories, affected mess halls, use cases, frequency, and student impact.

Tool 3: Food Complaint Task Analytics

Purpose: Provides operational information about complaint-related tasks.

Recommended output: open/resolved tasks, high-priority tasks, assignment groups, resolution status, task distribution, and pending workload.

Tool description: Analyzes food complaint tasks to identify priority, assignment, resolution status, workload, pending tasks, and operational areas requiring attention.

7. Create the Food Management Intelligence Agent

Agent Name: Food Management Intelligence Agent

Description:

The Food Management Intelligence Agent analyzes campus food service data to identify operational issues, recurring food problems, complaint trends, task status, priorities, and areas requiring attention.

The agent can use Food Management Analytics, Food Base Problem Analytics, and Food Complaint Task Analytics to retrieve relevant information from ServiceNow.

It should select appropriate analytics tools based on the user's request, combine information from multiple tools when required, identify meaningful patterns, and provide evidence-based insights and recommendations.

The agent must not invent data. All conclusions should be based on information returned by the available ServiceNow tools.

Agent Instructions

Analyze campus food management requests using the available ServiceNow analytics tools.

Understand the user's request before selecting tools.

Use Get Food Management Analytics for overall food management statistics and trends.

Use Food Base Problem Analytics for recurring food problems and problem categories.

Use Food Complaint Task Analytics for complaint task status, priority, assignment, workload, and resolution information.

Use multiple tools when the user's request requires information from multiple data sources.

Compare and interpret the retrieved information.

Identify important trends, high-priority issues, recurring problems, unresolved tasks, and areas requiring operational attention.

Provide recommendations based only on retrieved information.

Do not invent statistics, records, trends, or conclusions that are not supported by the available data.

Clearly distinguish between retrieved information, AI interpretation, and recommendations.

8. Add Tools to the Agent

- Get Food Management Analytics
- Food Base Problem Analytics
- Food Complaint Task Analytics

Verify that each tool is active, available to the agent, properly described, and returning usable information.

9. Test the AI Agent Independently

Test the agent before creating the Agentic Workflow. Start simple and gradually increase complexity.

Test 1:

Show me the current food management situation.

Test 2:

What are the most common food-related problems?

Test 3:

Which complaint tasks currently require attention?

Test 4:

Analyze the current food management situation using complaint, food problem, and task information. Identify the major issues and recommend actions.

Verify that the agent understands the request, selects the appropriate tools, retrieves data, combines results when needed, produces useful conclusions, and avoids unsupported assumptions.

10. Create the Agentic Workflow

Workflow Name: AI-Powered Campus Food Intelligence Workflow

Workflow Description:

The AI-Powered Campus Food Intelligence Workflow analyzes campus food service requests and coordinates the appropriate AI Agent and analytics capabilities to generate actionable operational insights.

When a user submits a request, the workflow should understand the required information, identify the appropriate AI Agent, provide the task to the agent, allow the agent to use relevant analytics tools, analyze the retrieved information, and return a structured response containing key findings and recommendations.

The workflow should use only available ServiceNow data and should not invent information that is not returned by the available tools.

11. Configure Workflow Steps

Step 1: Understand the Request

Understand the user's request and identify the campus food management information required to answer it.

Determine whether the request requires:

- Overall food management analytics

- Food base problem analysis
- Complaint task analysis
- Multiple sources of information

Only retrieve information relevant to the request.

Step 2: Select the AI Agent

Use the Food Management Intelligence Agent when the request requires analysis of campus food service data, complaints, recurring food problems, priorities, tasks, trends, or operational recommendations.

Provide the agent with the complete user request and relevant context.

Step 3: Analyze Relevant Data

Allow the selected AI Agent to use the appropriate analytics tools.

Use multiple tools when information from multiple data sources is required.

The agent should interpret the retrieved information and identify meaningful operational patterns.

Step 4: Generate Insights

Identify major issues, recurring problems, high-priority complaints, unresolved tasks, affected areas, and relevant operational trends based on the available data.

Do not create unsupported statistics or assumptions.

Step 5: Generate Recommendations

Provide practical recommendations based on the retrieved information.

Recommendations must be directly connected to the identified issues and should clearly distinguish AI recommendations from retrieved data.

Step 6: Return the Final Result

Return a structured response containing:

1. Executive Summary
2. Key Findings
3. Major Problem Areas
4. Priority or Operational Concerns
5. Recommended Actions
6. Supporting Data where relevant

12. Add the AI Agent to the Workflow

Add the Food Management Intelligence Agent to the workflow. Keep the terminology consistent between the workflow name, workflow description, workflow steps, AI Agent name, and AI Agent description.

Useful matching terms: Food Management, Food Intelligence, Food Analytics, Complaints, Food Problems, Complaint Tasks, Priorities, Operational Insights, Trends, Recommendations, Campus Food Services.

13. Use AI Agent Recommendations

- Open the AI Agent recommendation option.
- Ask Now Assist to recommend relevant AI Agents.
- Review the suggested agents.
- Confirm the Food Management Intelligence Agent is relevant.
- Add the appropriate agent to the workflow.

- Remove unrelated agents.

14. Primary End-to-End Test

Analyze the current campus food management situation.

Identify the most common food-related problems, major complaint categories, priority distribution, and current complaint task status.

Compare the food base problem information with complaint and task information to identify recurring issues that may require operational attention.

Highlight affected mess halls or meal types where the available data supports it.

Finally, provide the top three recommended actions for the food management team.

Use all relevant analytics tools available to you and clearly distinguish between retrieved data, identified trends, and recommendations.

15. Additional Test Cases

Overall Analytics

Give me an overview of the current campus food management situation.

Recurring Problems

What are the most common recurring food problems?

Operational Tasks

Which food complaint tasks require the most attention?

Multi-Source Intelligence

Identify the major food management problems and determine whether they are also reflected in complaint and task data.

16. What to Verify During Testing

Layer	Verify
Workflow	Understands request; identifies correct AI Agent; follows workflow instructions
Agent	Understands task; selects appropriate tools; combines results when needed
Tools	Return valid, relevant, understandable information
Analysis	Identifies useful patterns without unsupported assumptions
Response	Structured, clear, relevant, evidence-based recommendations

17. Demonstration Flow

1. Open the Agentic Workflow.
2. Show the workflow name, description, and steps.
3. Show the Food Management Intelligence Agent.
4. Show the three analytics tools.
5. Run the primary test request.
6. Open execution details / AI decision logs.
7. Show agent selection and tool invocation.
8. Show returned data.
9. Show the final AI-generated insights and recommendations.

18. AI Decision Logs

Decision or execution logs are valuable for demonstrating how the workflow moved from the user request to agent selection, tool invocation, analysis, and final response.

```
User Request
↓
Workflow Interpretation
↓
Agent Selection
↓
Tool Selection
↓
Tool Execution
↓
Data Analysis
↓
Final Response
```

19. Troubleshooting

Wrong AI Agent Selected

Make the agent description more specific and use consistent terminology across workflow and agent descriptions.

Required Tool Not Selected

Improve the tool description and explicitly state when the tool should be used. Example: use Food Complaint Task Analytics for complaint task status, priority, assignment, workload, or resolution.

Too Many Tools Selected

Tell the agent to use only tools relevant to the current request.

Unsupported AI Conclusions

Explicitly instruct the agent not to invent statistics, records, trends, or conclusions that are not supported by retrieved ServiceNow data.

Poor Final Response

Define a required output structure such as Executive Summary, Key Findings, Major Issues, Priority Concerns, Recommended Actions, and Supporting Data.

20. Best Practices

- Keep each tool specialized and focused on one clear capability.
- Keep the AI Agent focused on food management intelligence.
- Use precise descriptions because descriptions help the LLM match capabilities to workflow steps.
- Start with one AI Agent and a small set of useful tools before expanding.
- Use realistic application data.
- Test both simple and multi-tool requests.
- Keep recommendations grounded in retrieved enterprise data.
- Use human review for important operational decisions.

21. Security and Governance

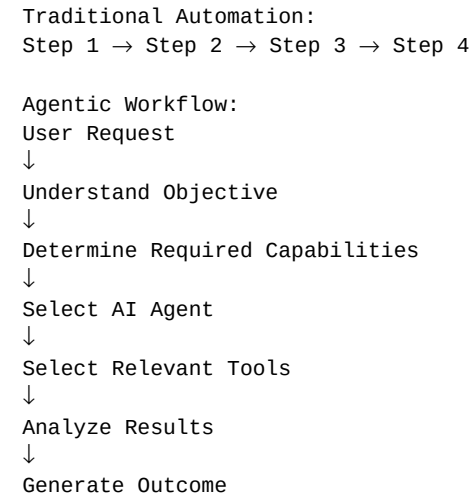
AI should not automatically receive unrestricted access to application data. Configure appropriate roles and access controls, expose only the data required by each tool, avoid unnecessary sensitive information, and treat AI-generated recommendations as decision support where appropriate.

22. Future Enhancements

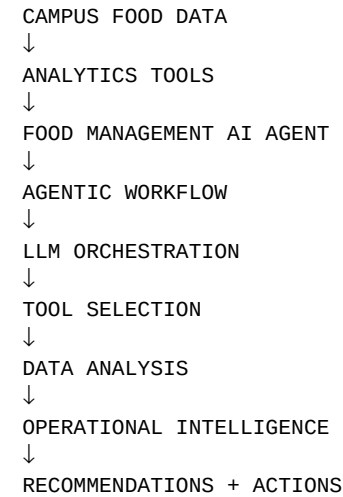
- Predictive insights based on historical trends.
- Automated notifications for critical patterns.
- Knowledge article integration.
- Role-based responses for students, staff, managers, and administrators.
- AI-generated insights integrated into dashboards.
- Human approval before important actions.
- Additional specialized agents for complaint, food quality, operations, or student feedback intelligence.

23. Why This Is an Agentic Workflow

A traditional automation follows a predefined sequence. An Agentic Workflow adds an AI orchestration layer that interprets the objective, identifies the required capability, selects relevant tools, analyzes results, and generates an outcome.



24. Final Architecture Summary



25. Quick Reference

Component	Configuration
Workflow	AI-Powered Campus Food Intelligence Workflow
AI Agent	Food Management Intelligence Agent
Tool 1	Get Food Management Analytics
Tool 2	Food Base Problem Analytics
Tool 3	Food Complaint Task Analytics
Core Flow	User Request → Agentic Workflow → AI Agent → Relevant Tools → AI Analysis → Insights → Recommendation
Platform	ServiceNow
Focus	AI Agents, Agentic Workflows, Enterprise AI, Operational Intelligence

26. Closing Note

This guide is intended as a practical reference for building the complete Agentic Workflow from the application data and AI tools through the AI Agent, workflow orchestration, testing, decision logs, and troubleshooting. The same architecture can be adapted to other enterprise use cases where an AI Agent needs to coordinate multiple tools and data sources.